

Name :- Y. Thanushma

Course Faculty :-  
Dr. R. Jeni Gracia

Register Number :- 192425355

Course Code :- ITA0502

CD-2 Assessment 3

24

25

1) Image Enhancement :-

Image enhancement in the spatial domain is the process of improving the visual quality of an image by directly modifying the pixel intensity values. It enhances contrast, removes noise and highlights important features. Spatial domain enhancement is represented as :-

$$g(x,y) = T[f(x,y)]$$

$f(x,y)$  = input image       $g(x,y)$  = enhanced image

a) Gray level Transformations :- These modify intensity of pixel values.

\* Image negative :- Enhances white details on dark background

$$S = L - 1 - r$$

\* log Transformation :- Expands darker pixels and compresses bright pixels.

$$S = c(\log(1+r))$$

\* Gamma Transformation :- Adjusts image brightness

$$S = Cr^{\gamma}$$

\* Contrast stretching :- Increase difference between dark and bright regions.

Suitable for :- Low contrast or purely illuminated images

Histogram Processing:- It represents the distribution of gray levels.

1. Histogram equalisation :- Improves overall contrast
  2. Histogram matching :- Produces a desired histogram shape.
  3. Histogram stretching :- Expands the intensity range
- Suitable for:- Images with poor contrast due to uneven lighting

Spatial Filtering:- Operates using a neighbourhood (mask / kernel)

1. Smoothing Filters (mean, Gaussian, median) :- Reduces Random noise.
2. Sharpening Filters (Laplacian, Sobel, Prewitt) :- Enhances edges and fine details.

Suitable for:- Noise removal and feature enhancement

~~Combining~~ Combining all these techniques significantly improves image quality for medical imaging, satellite images and surveillance systems.

②

| Image Smoothing         | Image Sharpening         |
|-------------------------|--------------------------|
| Reduces noise           | Enhances edges           |
| Removes fine details    | Highlights fine details  |
| uses low-pass filters   | use high-pass filters    |
| Produces blurred images | Produces clearer images. |

## Spatial Smoothing Filters:-

Replaces each pixel with average or weighted average of neighbouring pixels.

### Common filters:-

Mean filter  
Median filter  
Gaussian filter

### Mean Filter:-

$$g(x,y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} f(s,t)$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

### Advantages:-

- \* Removes Salt and Pepper noise
- \* Produces smooth images.

### Applications :-

- \* Medical Images
- \* Preprocessing before segmentation

### Sharpening Filters:-

- \* Detect rapid intensity changes (edges)
- \* Increase edge contrast.

### Common filters:-

Laplacian, Sobel, Perwitt

### Advantages:-

- \* Improves object boundaries
- \* Enhances image details.

### Applications :-

- \* Fingerprint recognition, Industrial Inspection, Medical diagnosis

Use Smoothing for noisy images

Use sharpening for blurred images requiring better edge visibility

③ Frequency Domain Enhancement:- Processes an image after converting it into the frequency domain using the Fourier Transform. Enhancement is achieved by multiplying the transformed image with a frequency filter and then applying the inverse transform.

Steps:-

1. Convert image using Fourier Transform
2. Apply Frequency Filter.  $G(u,v) = H(u,v) \times F(u,v)$
3. Perform Inverse Fourier Transform
4. Obtain enhanced image.

Low Pass Filter:- (LPF)

- \* Passes low frequency components
- \* Blocks high frequencies.
- \* Removes noise
- \* Produces smoother images. used before segmentation

Applications :-

Noise reduction, medical imaging, Satellite Image Enhancement.

High Pass Filter (HPF):-

- \* Passes high frequency components.
- \* Enhances edges.
- \* Improves fine details. used for feature extraction

Applications :-

Edge detection, object recognition, Industrial Quality Inspection.

~~Conclusion~~ - LPF is preferred for noise removal, whereas HPF is preferred for edge enhancement.

Na

Image Segmentation :- The process of dividing an image into meaningful regions to simplify analysis and object recognition.

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

| Technique                 | Working principle                            | Advantages                |
|---------------------------|----------------------------------------------|---------------------------|
| Point Detection           | Detects isolated pixels                      | Simple                    |
| Line Detection            | Detects horizontal, vertical, diagonal lines | Finds structural features |
| Edge detection            | Detects intensity changes                    | Accurate boundaries       |
| Thresholding              | Separates objects using intensity            | Fast and simple           |
| Region-based Segmentation | Groups similar neighbouring pixels           | Produces complete regions |

Limitations :-

- \* Sensitive to noise
- \* Missed curved lines
- \* Noise sensitive
- \* Poor under uneven lighting
- \* Computationally expensive

Most Suitable Technique :-

Region-based Segmentation

Reasons :-

- \* Produces accurate object boundaries
- \* Less affected by broken edges
- \* Suitable for medical imaging and object recognition

Applications :-

- \* Defect detection
- \* Road detection
- \* Medical imaging
- \* Document processing
- \* Medical image analysis

⑤ Edge Detection :-

Detects sudden changes in pixel intensity to identify object boundaries.

Working principle:-

Uses gradient operators such as Sobel, Prewitt, Canny, Laplacian.

gradient magnitude:-  
$$\sqrt{G_x^2 + G_y^2}$$

Applications :-

Face detection, medical imaging, industrial inspection

b) Edge linking:-

Connects broken or disconnected edges from continuous boundaries.

Working principle:-

Uses Hough Transform, Morphological operations contour tracing.

Applications :-

Shape Recognition, Road extraction, Character Recognition

c) Boundary Detection:-

Identifies the complete outer boundary of objects.

Working principle:-

uses connected edges and contour extraction algorithms

Applications :-

Object measurement, Defect detection, Image Segmentation

Ex:- Coin Inspection System

Edge Detection :- Detects the coin outline

Edge linking :- connects broken edge segments

Boundary Detection:- extracts complete circular boundary to measure diameter and detect defects.

Flow chart :-

Original Image  $\rightarrow$  Edge Detection  $\rightarrow$  Edge linking  
Defect detection  $\leftarrow$  Boundary detection  $\leftarrow$